

A Normal Stochastic Volatility Model

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Abstract

I derive a closed form formula for pricing a European style call option assuming stochastic volatility and normal dynamics for the underlying. The model differs from the Heston model in the underlying dynamics only – normal as opposed to lognormal.

1 Introduction

The drop in interest rates over the past few years to levels approaching, or sometimes below, zero has made it necessary to be able to price and hedge options with zero or negative strikes but without losing the ability to capture smiles and skews in quoted prices of options.

The Heston model [5] has the ability to fit skews and smiles. The associated closed form formula for the price of a vanilla option has made model calibration practical and thus enhanced the usefulness of the model by making it possible to price exotic options.

The Heston model generated a lot of interest since its publication. No attempt is made here at a comprehensive survey. We only mention that the contributions that ensued studied the numerical properties of the model (see, e.g., [2] and [8]), extensions of Heston's novel (to mathematical finance) approach (see, e.g., [3], [4], [9]) or extensions of the model to other dynamics (see, e.g., [1]). Where extensions have been studied, though, the dynamics of the underlying remained lognormal or didn't allow negative values of the underlying just as in Heston's original model.

The purpose of this paper is to introduce a closed form option pricing formula where the dynamics do allow the underlying to have negative values. It shares the same square-root process dynamics for the variance process with the Heston model but assumes normal, as opposed to lognormal, dynamics for the forward price process on the underlying.

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2 The Normal Model

We let x be the forward price process for an asset A . Then x is a martingale under the forward measure.¹ We assume that its variance process follows stochastic square-root dynamics. Specifically, our model is defined by the following stochastic differential equations:

$$\begin{aligned} dx &= \sqrt{\nu} dW \\ d\nu &= \kappa(\theta - \nu) dt + \omega\sqrt{\nu} dZ \\ dWdZ &= \rho dt \end{aligned} \quad (1)$$

where W and Z are Brownian motions under the forward measure with correlation ρ ; and κ , θ , and ω can be intuitively interpreted, respectively, as the speed of mean reversion of the variance, the long term limit variance and the volatility of the variance. This specification differs from Heston's in the first equation only where we adopt normal dynamics instead of Heston's lognormal. Also, the parameters of the model have essentially the same interpretation as in the Heston model except that here the variance is that of changes in levels.

Let c be the (forward) price of a call option on A struck at K and expiring in τ years, then, under the forward measure, we have²

$$c = E \left[(x_\tau - K)^+ \right]. \quad (2)$$

In the next section we will obtain a closed form formula for the Fourier transform of the probability density of x_τ . The goal in the rest of this section is to rewrite (2) in terms of that Fourier transform.

First we calculate the Fourier transform of the payoff function of the call option:³

$$\int_{-\infty}^{+\infty} e^{-i(-i\epsilon+\xi)x_\tau} (x_\tau - K)^+ dx_\tau = \frac{-e^{(-\epsilon-i\xi)K}}{(-i\epsilon + \xi)^2} \quad (3)$$

where we set the Fourier variable to $-i\epsilon + \xi$, with $\epsilon > 0$ and ξ a real number. The equation (3) holds $\forall \epsilon > 0$ and, crucially, the integral does not exist otherwise. The alternative approach where the payoff function is multiplied by a 'damping' factor prior to taking Fourier transforms, as in Carr and Madan [3], is equivalent to what we did here. Note that (3) is obtained using a simple integration by parts.

Finally, by definition, the Fourier transform ϕ of the probability density p of x at the option's expiry τ is

$$\phi(u) = \int_{-\infty}^{+\infty} e^{-iux_\tau} p(x_\tau) dx_\tau = E \left[e^{-iux_\tau} \right].$$

Note that $\phi(-u)$ is then the characteristic function of p . The call option price can be written in terms of ϕ and the Fourier transform of the payoff function courtesy of a

¹See, e.g., Jamshidian [6].

²With obvious notation: $(x_\tau - K)^+ = \max(x_\tau - K, 0)$.

³ $i = \sqrt{-1}$

judicious (and justified) use of Fubini's theorem. The appendix shows that for fixed $\epsilon > 0$, the following inversion formula holds:

$$(x_\tau - K)^+ = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{i(-i\epsilon + \xi)x_\tau} \frac{-e^{(-\epsilon - i\xi)K}}{(-i\epsilon + \xi)^2} d\xi. \quad (4)$$

Now we follow Carr and Madan [3] and Raible [9] and use this inversion formula to write the payoff function as the inverse transform of its own Fourier transform:

$$c = E[(x_\tau - K)^+] = E\left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(-i\epsilon + \xi)x_\tau} \frac{-e^{(-\epsilon - i\xi)K}}{(-i\epsilon + \xi)^2} d\xi\right].$$

Next rearrange the integrand and use Fubini to get

$$\begin{aligned} c &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} E\left[e^{i(-i\epsilon + \xi)x_\tau}\right] \frac{-e^{(-\epsilon - i\xi)K}}{(-i\epsilon + \xi)^2} d\xi \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \phi(i\epsilon - \xi) \frac{-e^{(-\epsilon - i\xi)K}}{(-i\epsilon + \xi)^2} d\xi. \end{aligned} \quad (5)$$

It can be verified that the integrand commutes with complex conjugation, so we get

$$c = \frac{1}{\pi} \int_0^\infty \Re\left[\phi(i\epsilon - \xi) \frac{-e^{(-\epsilon - i\xi)K}}{(-i\epsilon + \xi)^2}\right] d\xi \quad (6)$$

where $\Re$ is the real part of a complex number and again ϵ is a (small) fixed positive real number.

3 ϕ

To calculate ϕ , we use Fourier transforms to first find the solution to the backward Kolmogorov equation that governs the (joint) transition probability density of the system (1).

Let $P(t, x, \nu)$ be the probability density for the transition from x and ν to some terminal values (of the underlying and the variance) at time t . Then P satisfies the backward Kolmogorov equation⁴ for $t > 0$,

$$\frac{\partial P}{\partial t} = \kappa(\theta - \nu) \frac{\partial P}{\partial \nu} + \frac{1}{2} \nu \frac{\partial^2 P}{\partial x^2} + \frac{1}{2} \omega^2 \nu \frac{\partial^2 P}{\partial \nu^2} + \rho \omega \nu \frac{\partial^2 P}{\partial x \partial \nu}. \quad (7)$$

The parameters are as in the previous section. Let $\Phi(t, u, \nu)$ be the Fourier transform of P with respect to x , i.e.,

$$\Phi(t, u, \nu) = \int_{-\infty}^{\infty} e^{-iux} P(t, x, \nu) dx.$$

Then taking Fourier transforms in (7), we get

⁴See, e.g., Karatzas and Shreve [7].

$$\frac{\partial \Phi}{\partial t} = \kappa(\theta - \nu) \frac{\partial \Phi}{\partial \nu} - \frac{1}{2} \nu u^2 \Phi + \frac{1}{2} \omega^2 \nu \frac{\partial^2 \Phi}{\partial \nu^2} + \rho \omega \nu u i \frac{\partial \Phi}{\partial \nu}. \quad (8)$$

Now choose the Ansatz⁵

$$\Phi(t, u, \nu) = e^{C(t) + D(t)\nu - iux_0}.$$

Then substituting in (8) we get an expression that is identically 0 in ν and whose coefficients we therefore set to 0:

$$\dot{C} - \kappa\theta D = 0 \quad (9a)$$

$$\dot{D} - \frac{1}{2} \omega^2 D^2 + (\kappa - \rho \omega u i) D + \frac{1}{2} u^2 = 0 \quad (9b)$$

where dots denote differentiation with respect to t . If we fix the value of the variance at time 0, ν_0 , say, then $P(0, x, \nu_0)$ is the probability density of x at time 0 (conditional on ν_0) and we have

$$P(0, x, \nu_0) = \delta(x - x_0)$$

where δ is Dirac's delta function. $\Phi(0, u, \nu_0)$ is then its Fourier transform and we have

$$\Phi(0, u, \nu_0) = e^{-iux_0}.$$

Fixing x_0 and letting ν_0 vary, we conclude that we must have $D(0) = C(0) = 0$.

The Riccati equation (9b) for D can be solved by separating the terms in t and D and splitting the term in D into a sum of simple fractions. C is then obtained by integrating D with respect to t . We get

$$D(t) = \frac{\kappa - \rho \omega u i - s}{\omega^2} \left[\frac{1 - e^{-st}}{1 - ge^{-st}} \right]$$

$$C(t) = \frac{\kappa \theta}{\omega^2} \left[(\kappa - \rho \omega u i - s)t - 2 \log \left(\frac{ge^{-st} - 1}{g - 1} \right) \right]$$

and

$$g = \frac{\kappa - \rho \omega u i - s}{\kappa - \rho \omega u i + s}$$

$$s = \sqrt{(\kappa - \rho \omega u i)^2 + \omega^2 u^2}.$$

Finally, $\phi(u) = \Phi(\tau, u, \nu_0)$ is the Fourier transform of the probability density of the underlying at the option's expiry that we need in (6).

⁵Compare to Heston's.

4 Numerical Results

An extensive report on the numerical properties of the model and formula (6) will be the subject of a future note. The numbers below are to be treated as a simple illustration.

In these calculations, the integral in (6) was approximated using Gauss-Legendre quadrature (with the necessary variable transformation) using the nodes and weights from the statmod package in R and $\epsilon = 0.0001$. The Monte Carlo simulation of the system (1) did not include any of the usual refinements (variance reduction and such).

Table 1: Closed-form formula vs. Monte Carlo (MC). The option specs are: $x_0 = -10bps$; $\tau = 1$. The model parameters used are: $\kappa = 1$; $\theta = 0.5x_0^2$; $\omega = 0.25$; $\rho = -0.9$; $\nu_0 = 0.09$. The MC simulation used 100000 paths.

Strike	Formula	MC	Error
-0.0005	0.09223	0.09220	0.0003
0	0.09194	0.09197	0.0003
0.0005	0.09166	0.09152	0.0015

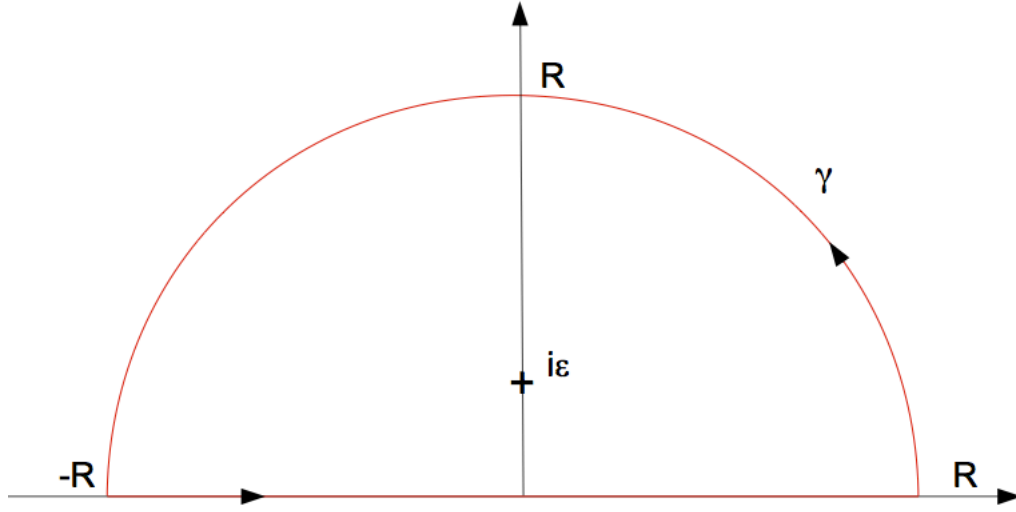


Figure 1: γ curve

Appendix: Inversion of the Fourier Transform of the Payoff Function

Here is a direct proof of the inversion formula (4) using elementary arguments.

Let γ be the curve in figure 1 travelled counter-clockwise and R the radius of the semi-circle. Let ϵ , such that $0 < \epsilon < R$, be given and

$$f(z) = e^{i(-i\epsilon+z)(x_\tau-K)}.$$

f as a function of a complex variable is then holomorphic over the whole plane (i.e., entire) and we can apply Cauchy's integral formula to the following contour integral:

$$\begin{aligned} \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(-i\epsilon+z)^2} dz &= \frac{df}{dz}(i\epsilon) \\ &= i(x_\tau - K). \end{aligned} \quad (10)$$

On the other hand,

$$\oint_{\gamma} \frac{f(z)}{(-i\epsilon+z)^2} dz = \int_{-R}^R \frac{f(z)}{(-i\epsilon+z)^2} dz + \oint_{\gamma_1} \frac{f(z)}{(-i\epsilon+z)^2} dz \quad (11)$$

where γ_1 is the semi-circle of radius R , i.e., $\gamma_1 = \gamma \setminus (-R, R)$. If $x_\tau - K > 0$, Jordan's lemma shows that the integral over γ_1 converges to 0 as $R \rightarrow +\infty$. Therefore,

$$x_\tau - K > 0 \implies \lim_{R \rightarrow +\infty} \frac{1}{2\pi i} \int_{-R}^R \frac{f(z)}{(-i\epsilon+z)^2} dz = i(x_\tau - K). \quad (12)$$

If $x_\tau - K < 0$, a similar argument that uses the contour integral along the reflection of γ with respect to the real axis leads to

$$x_\tau - K < 0 \implies \lim_{R \rightarrow +\infty} \frac{1}{2\pi i} \int_{-R}^R \frac{f(z)}{(-i\epsilon + z)^2} dz = 0. \quad (13)$$

If $x_\tau - K = 0$, $f(z)$ is identically 1 and we have

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{1}{(-i\epsilon + z)^2} dz = 0. \quad (14)$$

Moreover, along γ_1 ,

$$\oint_{\gamma_1} \frac{1}{|-i\epsilon + z|^2} dz \leq \frac{\pi R}{(R - \epsilon)^2}. \quad (15)$$

Therefore we have

$$x_\tau - K = 0 \implies \lim_{R \rightarrow +\infty} \frac{1}{2\pi i} \int_{-R}^R \frac{f(z)}{(-i\epsilon + z)^2} dz = 0. \quad (16)$$

We get (4) by combining (12), (13) and (16).

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